

4. MINIMIZING THE RATE FUNCTION ON THE BOUNDARY

From now on we will work exclusively with graphons rather than with graphs. From Theorem 3.1, all questions boil down to “minimize the rate function over such-and-such region”. The first region we study is the lower boundary of (e, t) -space, beginning with the first (flat) scallop:

Theorem 4.1. *If $e \leq 1/2$ and $t = 0$, then $\min_{F_0} I(g) = I_0(2e)/2$, and this minimum is achieved at the graphon*

$$(13) \quad g_0(x, y) = \begin{cases} 2e & \text{if } x < \frac{1}{2} < y \text{ or } y < \frac{1}{2} < x; \\ 0 & \text{otherwise.} \end{cases}$$

Furthermore, any other minimizer is equivalent to g_0 , corresponding to the same reduced graphon.

Proof. Since $t(g)$ is identically zero, the measure of the set $\{(x, y) \in [0, 1]^2 | g(x, y) = 0\}$ is at least $1/2$. Otherwise, the graphon $\bar{g}(x, y) = \begin{cases} 1 & \text{if } g(x, y) > 0; \\ 0 & \text{otherwise,} \end{cases}$ would have no triangles and would have edge density greater than $1/2$, which is impossible. So we restrict attention to graphons that are zero on a set of measure at least $1/2$ and have edge density e . From the convexity of I_0 , we know that the graphon minimizing I must be zero on a set of measure $1/2$ and must be constant on the rest. Thus g_0 is a minimizer, and $\min_{F_0} I(g) = I(g_0) = I_0(2e)/2$.

Now suppose that g is another minimizer. Since g is zero on a set of measure $1/2$ and is $2e$ on a set of measure $1/2$, $\bar{g}$ is 1 on a set of measure $1/2$, and so describes a graphon with edge density $1/2$ and no triangles. This means that $\bar{g}$ describes a complete bipartite graph with the two parts having the same measure. That is, $\bar{g}$ is equivalent to the graphon that equals 1 if $x < \frac{1}{2} < y$ or $y < \frac{1}{2} < x$ and is zero everywhere else. But then $g = 2e\bar{g}$ is equivalent to g_0 . $\square$

The situation on the curved scallops is slightly more complicated. Pick an integer $\ell > 1$. (The case $\ell = 1$ just gives us our first scallop.) If $e \in \left[1 - \frac{1}{\ell}, 1 - \frac{1}{\ell+1}\right]$, then any graph G with edge density e and the minimum number of triangles has to take the following form (see [PR] for the history). Let

$$(14) \quad c = \frac{\ell + \sqrt{\ell(\ell - e(\ell + 1))}}{\ell(\ell + 1)}.$$

There is a partition of $\{1, \dots, n\}$ into ℓ pieces, the first $\ell - 1$ of size $\lfloor cn \rfloor$ and the last of size between $\lfloor cn \rfloor$ and $2\lfloor cn \rfloor$, such that G is the complete ℓ -partite graph on these pieces, plus a number of additional edges within the last piece. ($\lfloor y \rfloor$ denotes the largest integer greater than or equal to y .) These additional edges can take any form, as long as there are no triangles within the last piece.

This means that, after possibly renumbering the vertices, the graphon for such a graph can be written as an uneven $\ell \times \ell$ checkerboard obtained from cutting the unit interval into pieces $V_k = [(k-1)c, kc]$ for $k < \ell$ and $V_\ell = [(\ell-1)c, 1]$, with the checkerboard being 1 outside the

main diagonal, 0 on the main diagonal except the upper right corner, and corresponding to a zero-triangle graph in the upper right corner.

Limits of such graphons in the metric must take the form

$$(15) \quad g(x, y) = \begin{cases} 1 & x < kc < y \text{ or } y < kc < x \text{ for an integer } k < \ell; \\ 0 & (k-1)c < x, y < kc \text{ for some integer } k < \ell; \\ \text{unspecified} & x, y > (\ell-1)c, \end{cases}$$

with

$$(16) \quad \iiint_{[(\ell-1)c, 1]^3} g(x, y)g(y, z)g(z, x) dx dy dz = 0,$$

and with $\iint_{[0, 1]^2} g(x, y) dx dy = e$. Minimizing $I(g)$ on such graphons is easy, since all but the upper right corner of the graphon is fixed. Applying Theorem 4.1 to that corner, we get

Theorem 4.2. *If $e > 1/2$ and t is the smallest value possible, then the minimum of $I(g)$ on F_0 is achieved by the graphon*

$$(17) \quad g_0(x, y) = \begin{cases} 1 & x < kc < y \text{ or } y < kc < x \text{ for an integer } k < \ell; \\ p & (\ell-1)c < x < [1 + (\ell-1)c]/2 < y \text{ or } (\ell-1)c < y < [1 + (\ell-1)c]/2 < x; \\ 0 & \text{otherwise,} \end{cases}$$

where

$$(18) \quad p = \frac{4c(1 - \ell c)}{(1 - (\ell-1)c)^2}$$

is a number chosen to make $\iint_{[0, 1]^2} g(x, y) dx dy = e$. Furthermore, any other minimizer is equivalent to g_0 . The minimum value of $I(g)$ is

$$(19) \quad I(g_0) = \frac{(1 - (\ell-1)c)^2}{2} I_0(p).$$

5. MINIMIZING NEAR THE FIRST SCALLOP

Now that we know the minimizer at the (lower) boundary, we perturb it to get a minimizer near the boundary.

Theorem 5.1. *Pick $e < 1/2$ and ϵ sufficiently small. Then the graphon*

$$(20) \quad g(x, y) = \begin{cases} 2e - \epsilon & x < \frac{1}{2} < y \text{ or } y < \frac{1}{2} < x \\ \epsilon & \text{otherwise,} \end{cases}$$

minimizes the rate function to second order in perturbation theory among graphons with $e(g) = e$ and $t(g) = e^3 - (e - \epsilon)^3$. For pointwise small variations δg of g , the second variation in $I(g)$ is bounded from below by $\frac{1}{2} \iint_{[0, 1]^2} (\delta g(x, y))^2 dx dy$.